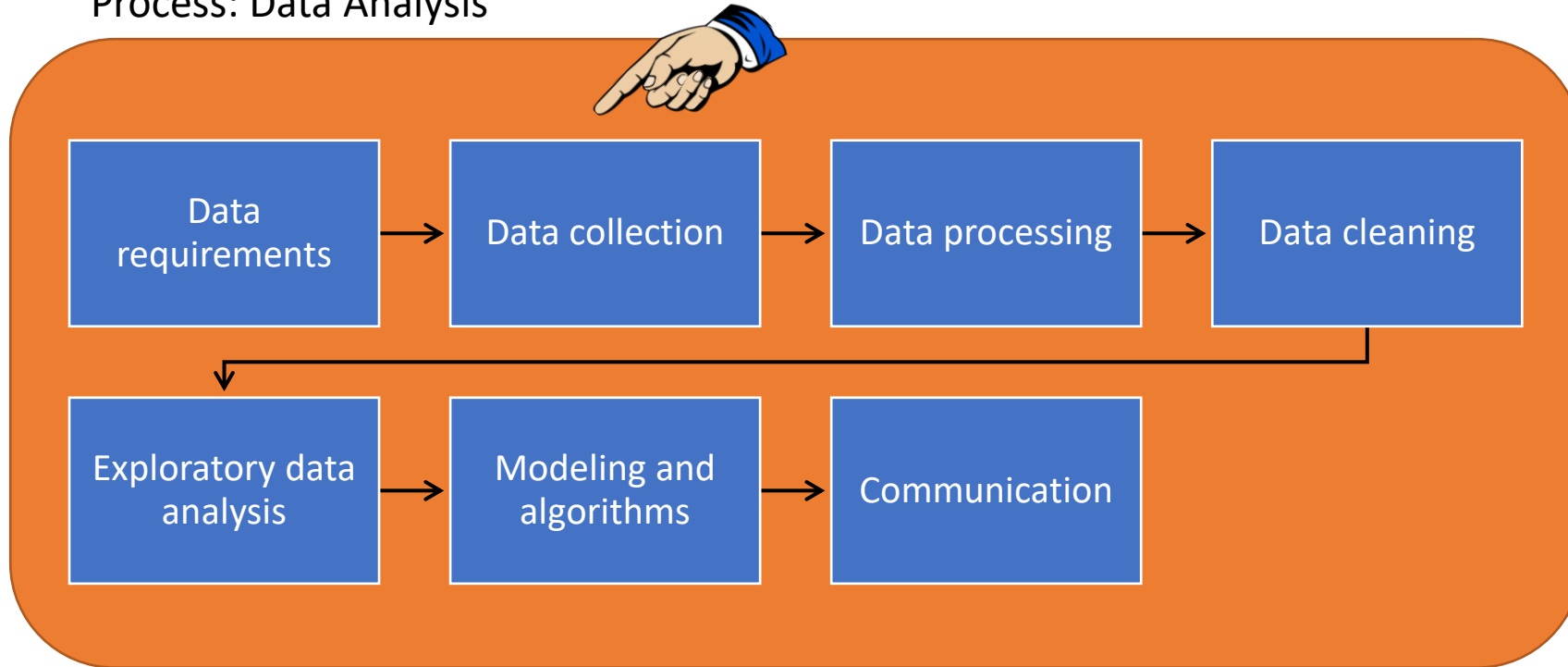


Data Collection



Process: Data Analysis



Tools

Excel



Zotero



Trello



Methods



Considerations



Techniques

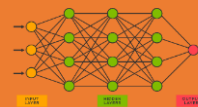


Descriptive Analytics

Diagnostic Analytics

Predictive Analytics

Prescriptive Analytics



Learning Outcomes for Data Collection

You will learn about:

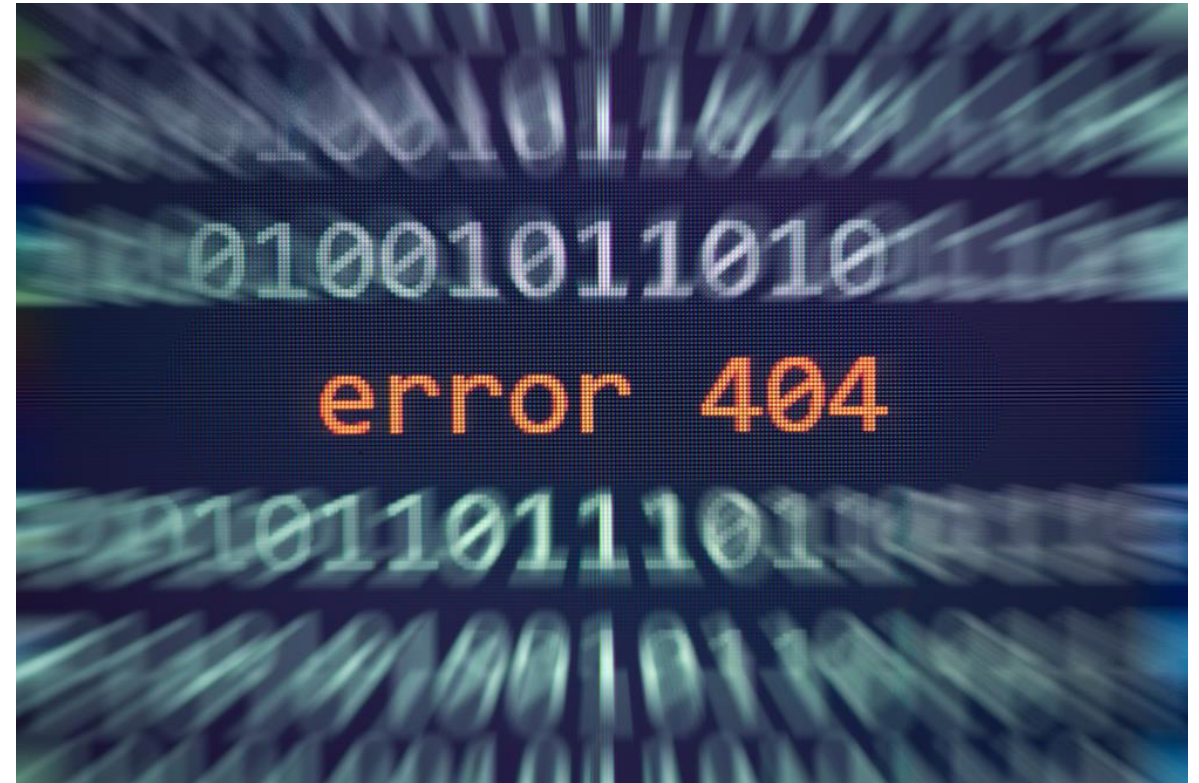
- The importance of strategy when engaging in data projects.
- Types of Data
- Structured and Unstructured Data.
- Primary Data
- Secondary Data
- How to use Search Engines to find information.

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1. The importance of strategy when engaging in data projects. 
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Data Projects are hard!

- 85% of big data projects fail ([Gartner](#), 2017)
- 87% of data science projects never make it to production ([VentureBeat](#), 2019)
- “Through 2022, only 20% of analytic insights will deliver business outcomes” ([Gartner](#), 2019)



The importance of strategy when engaging in data projects.

- In [McKinsey](#)'s work with companies in six data-rich industries, they found that fully exploiting data and analytics requires [three mutually supportive capabilities](#).

The McKinsey & Company logo is displayed in white serif font on a dark blue rectangular background. The text is arranged in two lines: "McKinsey" on the top line and "& Company" on the bottom line, with the ampersand being particularly large and stylized.

McKinsey
& Company

Three keys to building a data-driven strategy

1. **Data:** Companies must be able to identify, combine, and manage **multiple sources of data**.
2. **Modeling:** Companies need the capability to build advanced-analytics models for **predicting** and **optimizing outcomes**.
3. **Implementation:** Management must possess the muscle to **transform the organization** so that the data and models yield better decisions.



Choose the right data

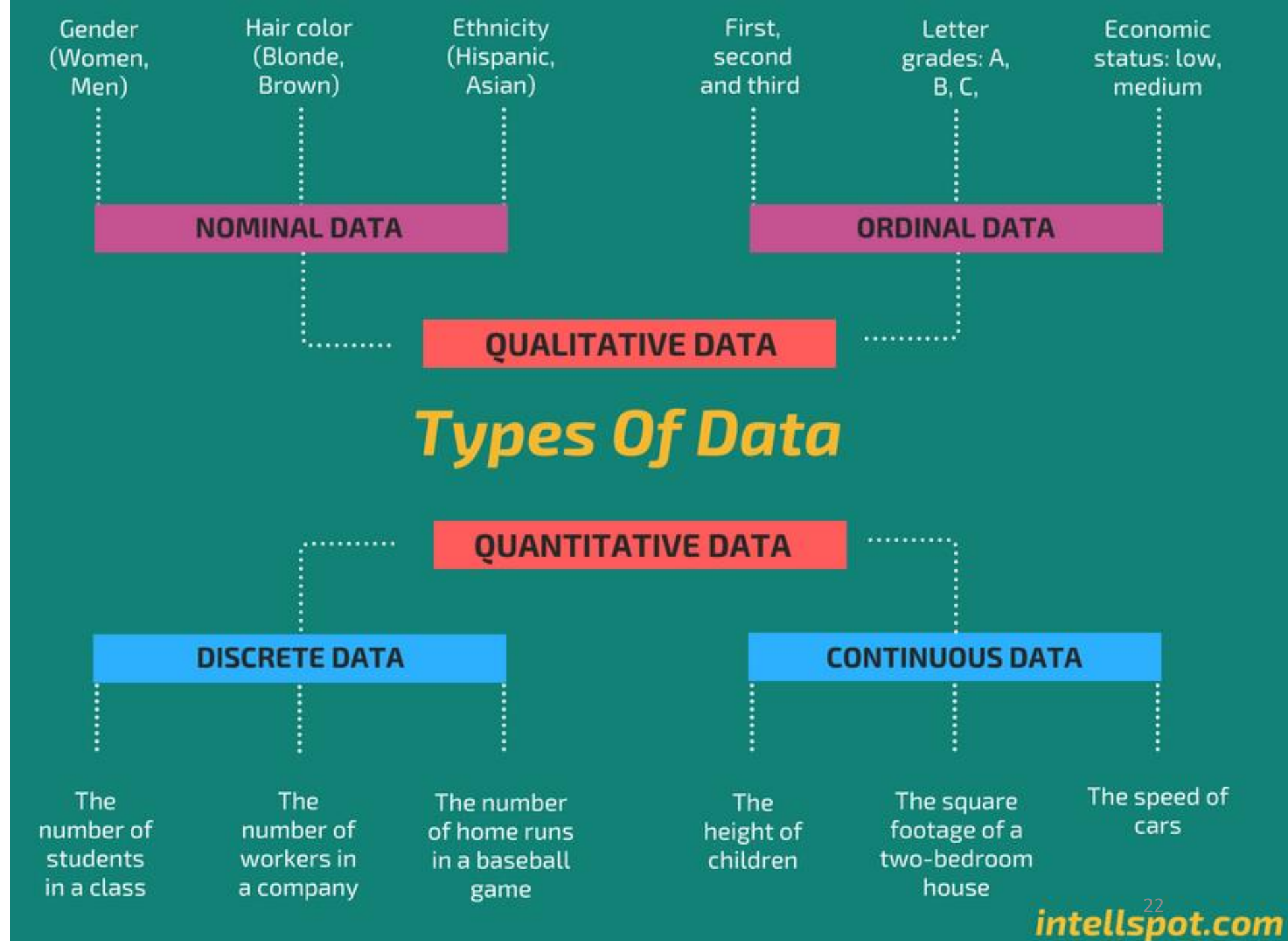
- **Accuracy:** Ensures trustworthy results.
- **Relevance:** Aligns with project goals.
- **Completeness:** Prevents gaps in analysis.
- **Consistency:** Avoids discrepancies.
- **Bias Reduction:** Minimizes skewed findings.
- **Efficiency:** Saves time and resources.
- **Better Models:** Improves predictive accuracy.
- **Compliance:** Meets legal standards.
- **Scalability:** Supports future needs.
- **Actionable Insights:** Drives decisions.



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“Traditional” Data categories



Qualitative Data

- Qualitative data is non-numeric* information that describes characteristics, qualities, or attributes of a subject.
- Can be Nominal or Ordinal.



*Sometimes numbers can be used to represent qualitative data, but they only represent labels rather than actual quantities

Nominal Qualitative Data

- Used just for labeling variables, without any type of quantitative value.
- There is no intrinsic ordering to the variables.
- Examples of Nominal data: nationality, gender, hair color.



Ordinal Qualitative Data

- **Ordered Categories:** Data is arranged in a specific order or ranking on an arbitrary scale.
- **Relative Position:** The order matters, but the differences between positions are not consistent or measurable.
- **Assigned Numbers:** Numbers can be used to indicate rank, but they don't represent actual quantities.
- **No Mathematical Operations:** Arithmetic operations (e.g., addition, subtraction) are not meaningful with these numbers.
- **Example:** Olympic medals (Gold, Silver, Bronze) are ranked, but Gold isn't quantitatively more than Silver.



Nominal Qualitative Data (No Order):

- **Genres of Music:** Jazz, Rock, Classical, Hip-Hop.
- **Types of Cuisine:** Italian, Thai, Mexican, Indian.
- **Product Categories:** Electronics, Clothing, Furniture, Books.
- **Types of Transportation:** Car, Bicycle, Bus, Train.
- **Eye Colors:** Blue, Green, Brown, Hazel.

Ordinal Qualitative Data (Ordered):

- **Education Levels:** High School, Bachelor's, Master's, Doctorate.
- **Customer Satisfaction Ratings:** Very Dissatisfied, Dissatisfied, Neutral, Satisfied, Very Satisfied.
- **Pain Intensity Levels:** No Pain, Mild, Moderate, Severe, Extreme.
- **Spice Levels in Food:** Mild, Medium, Hot, Extra Hot.
- **Performance Reviews:** Needs Improvement, Meets Expectations, Exceeds Expectations.

Nominal vs Ordinal Data



Nominal Data

Nominal data is used just for labeling variables, without any type of quantitative value. The name 'Nominal' comes from the Latin word "nomen" which means 'name'.

The nominal data just name a thing without applying it to an order. Actually, the nominal data could just be called "labels."

Ordinal data

Ordinal data is data which is placed into some kind of order by their position on a scale.

Ordinal data may indicate superiority. However, you cannot do arithmetic with ordinal numbers because they only show sequence.

Examples

- Gender (Women, Men)
- Hair color (Blonde, Brown, Brunette, Red, etc.)
- Marital status (Married, Single, Widowed)
- Ethnicity (Hispanic, Asian)



Examples

- The first, second and third person in a competition.
- Letter grades: A, B, C, and etc.
- When a company asks a customer to rate the sales experience on a scale of 1-10.
- Economic status: low, medium and high.



Quantitative Data

- Data that can be expressed as a number and measured.
- Can be Discrete or Continuous.



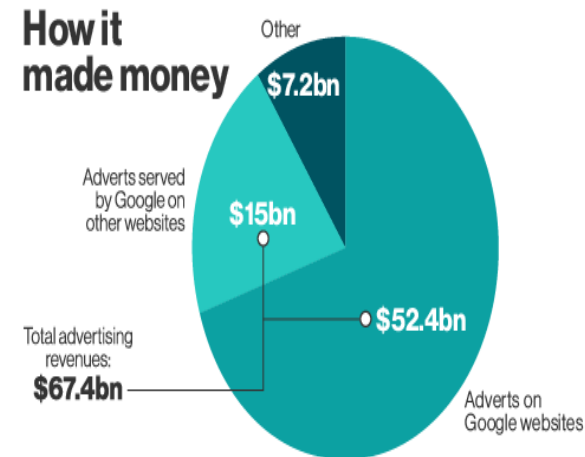
Discrete Quantitative Data

- Discrete data is a count that involves only integers.
- The discrete values cannot be subdivided into smaller parts.
- **Examples:** number of students in a class; number of goals in a football match; days of the month...



Continuous Quantitative Data

- Data that can be divided into finer levels.
- Includes any number within a certain range.
- It can take almost any numeric value.
- **Examples:** income; weight; height; temperature; time...



Data Collection Table of Contents

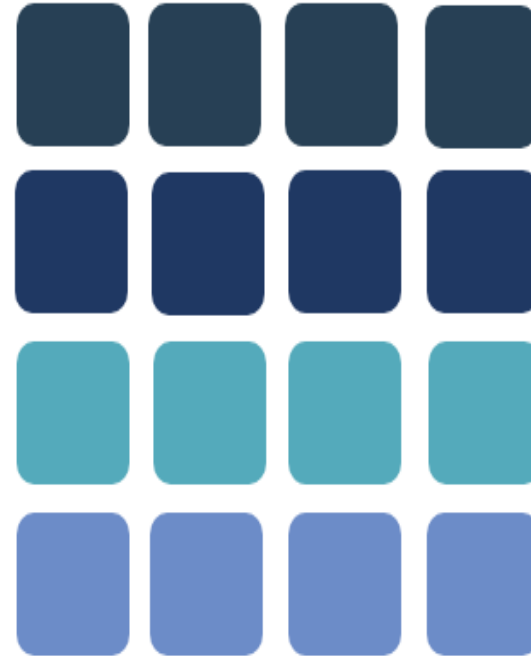
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A new way of looking at types of data

UNSTRUCTURED DATA

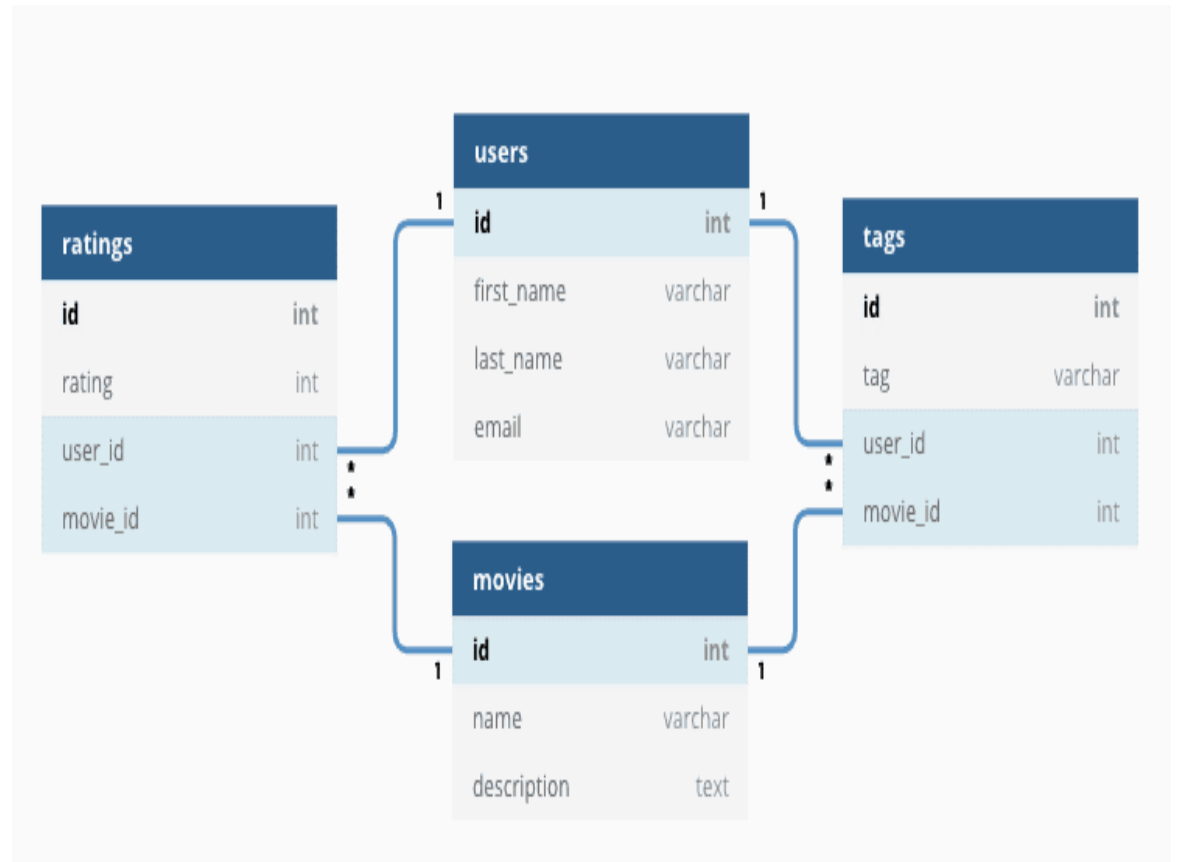
VS

STRUCTURED DATA



What is Structured Data?

- **Organized Format:** arranged in rows and columns (e.g., databases, spreadsheets); stored in relational databases.
- **Rule of thumb:** if you can use Excel with it, your Data is Structured.
- **SQL:** Structured Query Language (SQL) is the programming language used to manage structured data.
- **Nature of the data:** It can store both quantitative and qualitative data, although it is usually akin to quantitative data
- **Examples:** Names, dates, sales figures.



Pros of Structured Data

- **Easy to Work With:** Structured data's organized format makes it simple to manipulate, query, and analyze with standard tools.
- **User-Friendly:** Accessible to business users with basic knowledge, allowing for easy access and interpretation.
- **Wide Tool Support:** More tools are available for structured data analysis due to its long-standing use in data management.



Cons of Structured Data

1. **Limited usage:** Data with a predefined structure can only be used for its intended purpose, which limits its flexibility and usability.
2. **Rigid storage options:** Generally stored in systems with rigid schemas (e.g., “data warehouses”), making it inflexible to changes or additions without redesigning the structure, which leads to massive expenditures of time and resources.
3. **High Maintenance cost:** Keeping the data clean, consistent, and up-to-date requires ongoing maintenance and data management.
4. **Limited Insight:** May miss valuable insights that can be derived from unstructured data, such as sentiment or behavioral patterns.



Example of uses of Structured Data

Customer Relationship Management (CRM)

- **CRM Systems:** Use structured data like customer details, purchase history, and interaction logs, organized into fields within databases.
- **Data Analysis:** Analyzes data to identify customer behavior patterns and trends.
- **Key Insights:** Reveals top products, customer lifetime value, and service needs.
- **Business Impact:** Helps tailor marketing, improve service, and boost sales.



Examples of uses of Structured Data

Online booking

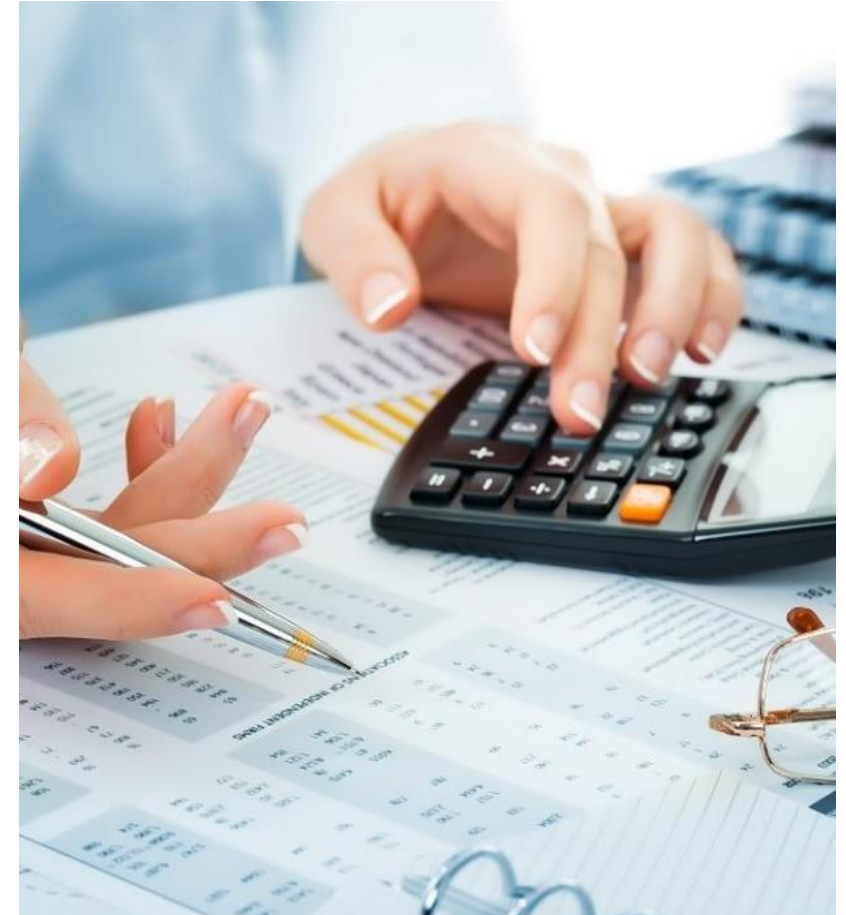
- **Online Booking Systems:** Use structured data like dates, prices, destinations, and customer details.
- **Organized Format:** Data fits into rows and columns, making it easy to search and filter.
- **Efficiency:** Quickly checks availability, manages inventory, and calculates costs.
- **User Experience:** Provides accurate, fast, real-time booking information.



Examples of uses of Structured Data

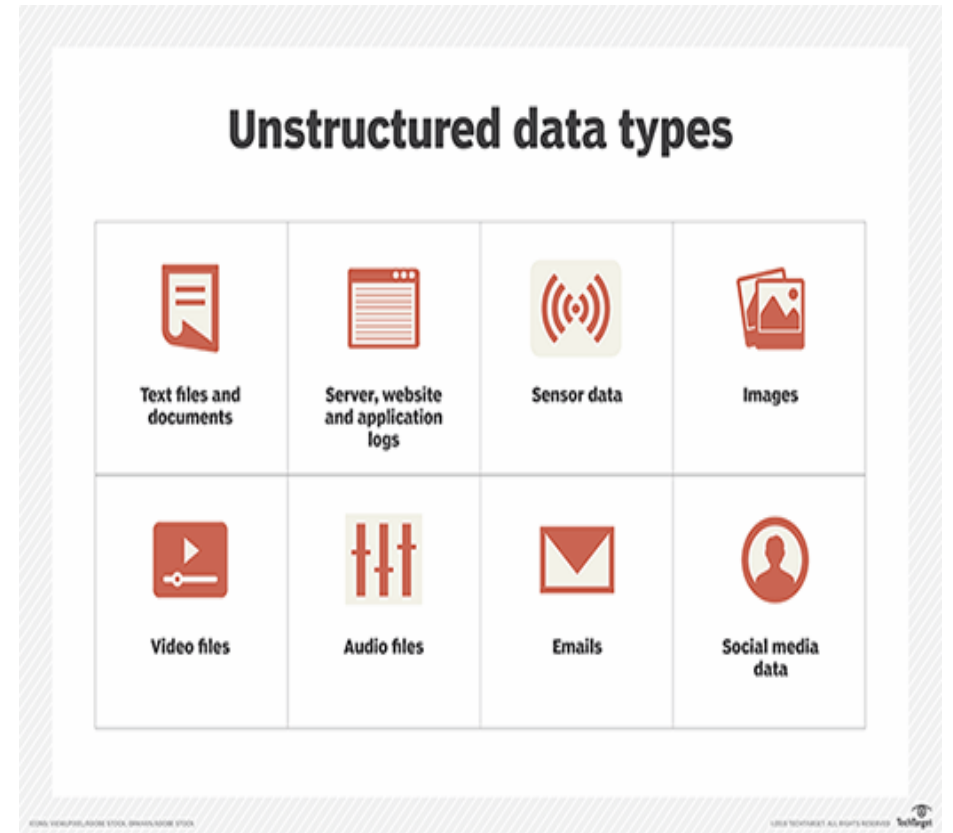
Accounting

- **Accounting Use:** Structured data includes invoices, payments, expenses, and account balances.
- **Organized Storage:** Data is kept in rows and columns in accounting software.
- **Efficiency:** Enables easy tracking, auditing, and reporting of financial transactions.
- **Automation:** Supports automated calculations for statements, reconciliations, and cash flow analysis.
- **Accuracy and Compliance:** Reduces errors and ensures compliance with financial regulations.



Unstructured Data

- Cannot be processed and analyzed via conventional data tools and methods.
- Typically categorized as qualitative data.
- Is **stored** in its **native format** and not processed until it is used.
- Their benefits involve **advantages in format, speed and storage**, while limitations revolve around the **required expertise** and **limited availability of tools**.
- **Rule of thumb:** any data you cannot use in Excel.



Examples of Unstructured Data



Communications

All types of communication such as live chat, collaboration software, IM, etc.



Customer-generated content

Online comments, emails or calls to customer service department, etc.



Raw data from research

Surveys, focus groups, interviews are common unstructured data examples



Books, magazines, blogs

All of the writings (books, news, articles, press releases, and etc.)



Medical records

The significant list of healthcare records that create the whole “patient picture”



Scientific data

Oil gas exploration, seismic imagery, atmospheric data, space exploration and so on.

Examples of Unstructured Data

Source: [Examples of Unstructured Data @ Intellspot](#)



Emails

Among the most popular unstructured data we use every day for business



Text files

Word processing, spreadsheets, PDF files, reports, graphics files, XML files and etc....



Websites

YouTube, photo sharing sites, Instagram, Flickr, business websites, etc...



Social Media

The data generated from the social media platforms such as Facebook, Twitter, LinkedIn



Media

All sorts of media such as digital images, audio, video, MP3, and etc.



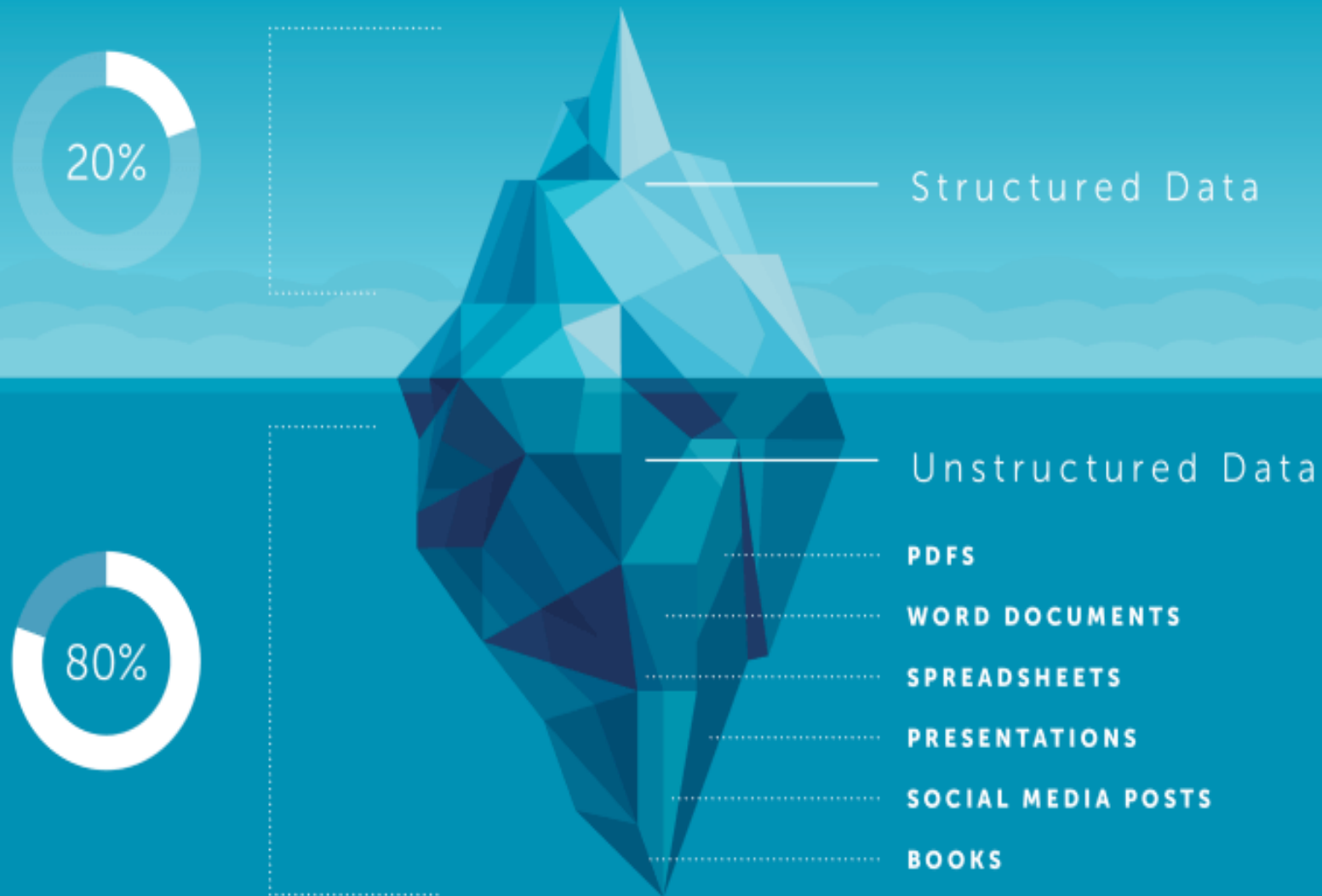
Mobile data

This includes information such as different types of text messages and location

It's huge, it's everywhere, it's here!

- Unstructured data represents about 80% of all new data.
- It's growing three times faster than structured data.
- [IDC predicts](#) that the volume of data is set to grow from 33 zettabytes in 2018 to 175 zettabytes (175 billion terabytes!) by 2025

Source: [Possibilities and limitations, of unstructured data](#)



Pros of Unstructured Data

1. **Volume and Variety:** : Large volumes of diverse data sources (social media, emails).
2. **Rich Insights:** Captures detailed information like emotions, opinions, and behaviors.
3. **Flexibility:** No rigid format, allowing for a wide range of data types (text, images, videos); stored in its native format, remains undefined until needed
4. **Real-World Context:** Provides context and depth that structured data often misses.
5. **Enhanced Analytics:** Supports advanced analytics like sentiment analysis and trend spotting.



Cons of Unstructured Data

1. **Requires expertise:** Due to its undefined/non-formatted nature, data science expertise is required to prepare and analyze unstructured data. This alienates unspecialized business users.
2. **Specialized tools:** Specialized tools are required to manipulate unstructured data, which limits product choices for data managers.
3. **Expensive Storage:** Needs more storage space and specialized systems to manage



Examples of uses of Unstructured Data

- **Social Media Monitoring:** Analyzing posts, comments, and mentions to understand audience sentiment, track brand perception, and optimize content strategies for increased engagement and follower growth.
- **Customer Support:** Mining chat logs, emails, and call transcripts to improve customer service and identify common issues.
- **Content Personalization:** Using browsing history and user-generated content to personalize recommendations and advertisements.
- **Market Research:** Analyzing product reviews and forums to understand consumer opinions and identify market trends.
- **Healthcare:** Extracting information from medical records, doctor's notes, and imaging data for diagnostics and patient care.
- **Recruitment:** Scanning resumes, cover letters, and social profiles to identify suitable job candidates.

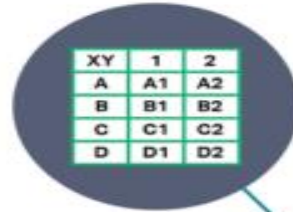


Structured Data

VS

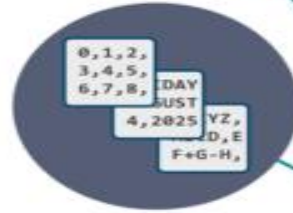
Unstructured Data

Can be displayed
in rows, columns and
relational databases



XY	1	2
A	A1	A2
B	B1	B2
C	C1	C2
D	D1	D2

Numbers, dates
and strings



Estimated 20% of
enterprise data (Gartner)



Requires less storage



Easier to manage
and protect with
legacy solutions



Cannot be displayed
in rows, columns and
relational databases



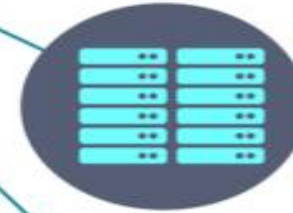
Images, audio, video,
word processing files,
e-mails, spreadsheets



Estimated 80% of
enterprise data (Gartner)



Requires more storage



More difficult to
manage and protect
with legacy solutions



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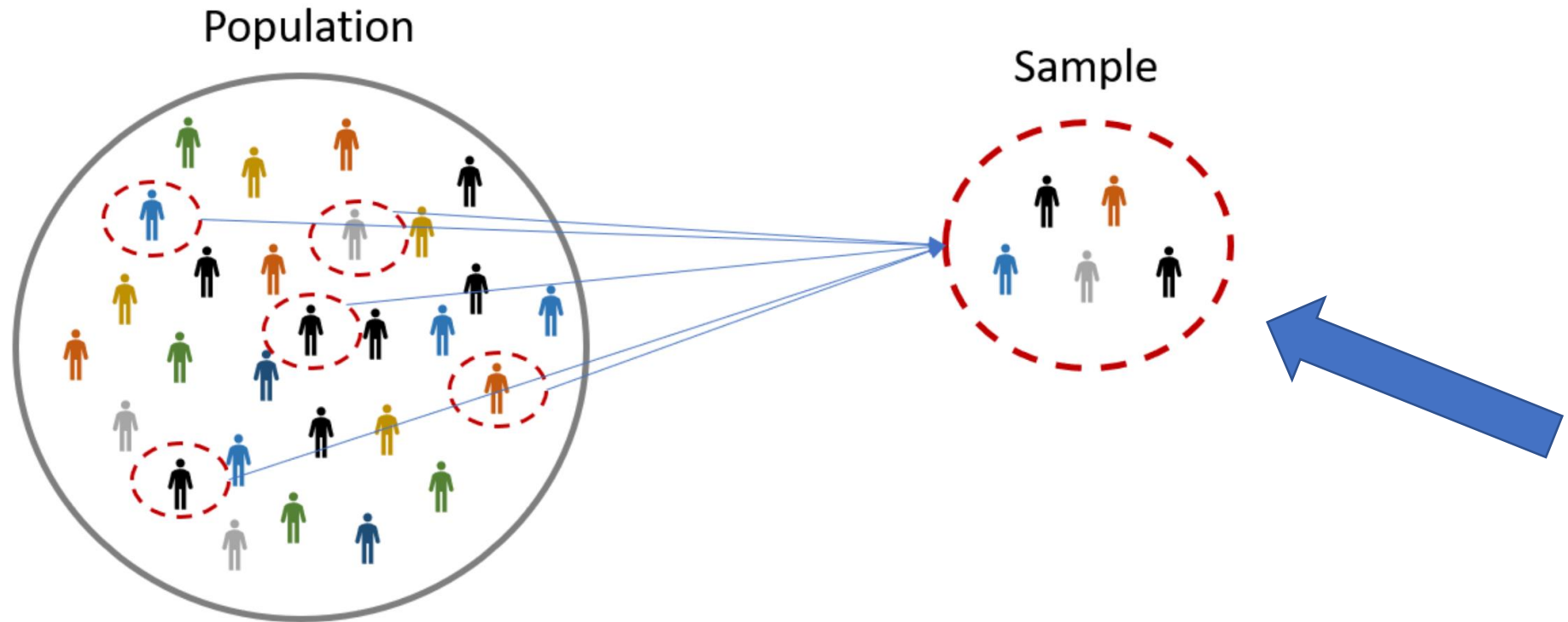
What is Primary Data

- Primary data is information collected **directly** from first-hand experience.
- This is the (first-hand) data that **the researcher gathers** for the purpose of a **particular project**.



Sources of Primary Data

- The source of primary data is the population sample from which you gather your data.



Methods of obtaining Primary Data

- Experiments
- Interview
(personal interview, telephone, e-mail)
- Self-administered surveys and questionnaires
- Field observation
- Case studies
- Diary entries, letters, and other correspondence
- Eyewitness accounts
- Ethnographic research
- Personal narratives, memoirs



Advantages of Primary Data

- **Addresses Specific Issues:** Primary data directly tackles the specific needs and questions of your business, offering tailored insights not available through secondary data.
- **High Accuracy:** Data is collected firsthand, ensuring it accurately reflects the target population without biases inherent in secondary sources.
- **Greater Control:** You have full control over the research design, methodology, and data collection process, ensuring it aligns perfectly with your objectives, and minimizing common issues like bias and missing values.
- **Up-to-Date Information:** Primary data provides the most current insights as it is collected in real-time from the field, unlike often outdated secondary data.
- **Exclusive Ownership:** You own the collected data, giving you a competitive edge as it is unique and not shared with competitors.



Disadvantages of Primary Data

- **High Cost:** Primary data collection is often expensive as it requires starting from scratch, including designing the study, organizing materials, and conducting the entire research process.
- **Time-Consuming:** Conducting primary research takes significantly more time compared to gathering secondary data, from planning to data collection and analysis.
- **Limited Scope:** Primary data is often restricted by specific time frames, locations, or sample sizes, offering a narrower perspective compared to the broader range available from secondary sources.
- **Feasibility Challenges:** Some studies may be too large, complex, or resource-intensive for your organization to carry out effectively.



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Secondary Data

- **Second-hand** information: collected by others than the researchers themselves
- Data that have been **already collected for another purpose** but has some **relevance** to your research needs.



Examples of secondary data sources

- **Previous Research:** Reports from earlier research efforts relevant to your topic.
- **Journal Articles:** Academic publications offering peer-reviewed findings.
- **Dissertations and Theses:** In-depth research work by students, often containing original data and comprehensive literature reviews.
- **Government Reports:** Policy documents, white papers, public records.
- **Official Statistics:** Census data, economic indicators.
- **Web Information:** Websites, blogs, digital archives.
- **Historical Data:** Past records, historical documents, previous datasets.
- **Monographs:** Books and papers on specific subjects.
- **Financial Reports:** Company financial statements, market analyses.

MIS
Quarterly

SPECIAL ISSUE

DIGITIZATION AND PHASE TRANSITIONS IN PLATFORM ORGANIZING LOGICS: EVIDENCE FROM THE PROCESS AUTOMATION INDUSTRY¹

Johan Sandberg and Jonny Holmström

Swedish Center for Digital Innovation, Department of Informatics, Umeå University,
87 Umeå, SWEDEN {johan.sandberg@umu.se} {jonny.holmstrom@umu.se}

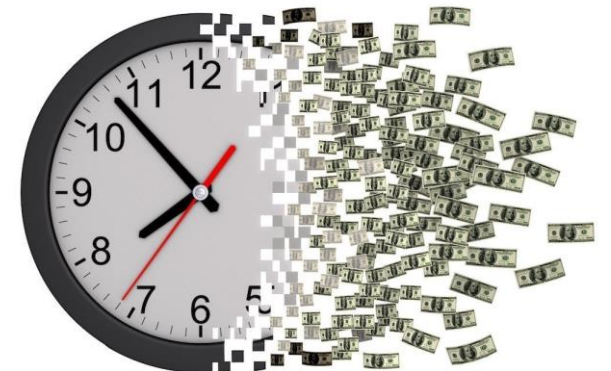
Kalle Lyytinen

Weatherhead School of Management, Case Western Reserve University,
Cleveland, OH 44106 U.S.A. {kalle@case.edu}

This paper draws on complex adaptive systems (CAS) theory to explore the transformation of an analog automation product platform as it was infused with extensive and deepening digital capacities over a 40-year period. Our case demonstrates how the deepening digitization of components and functions drives complexity by connecting the platform to multiple social and technical settings and producing new interactions and information exchanges. The increased connectivity and dynamism invited unexpected and significant architectural and organizational shifts that moved the platform toward an ecosystem-centered organizing logic. CAS theory and its notion of constrained generating procedures (CGPs) are used to analyze how new connections

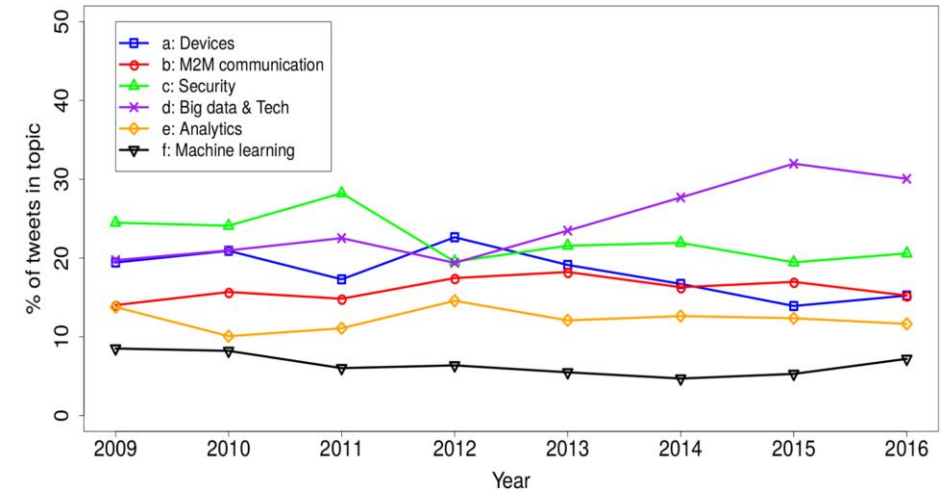
Advantages of Secondary Data

- **Ease of Access:** Secondary data is readily accessible online or through libraries and databases, often requiring just a few clicks to obtain.
- **Low Cost or Free:** Many secondary sources are available at little to no cost, saving money and reducing the effort needed for data collection.
- **Time-Saving:** Secondary research can be completed quickly, often with simple searches, allowing for faster insights without the lengthy process of primary data collection.



Advantages of Secondary Data

- **Longitudinal Analysis:** Allows for analysis over extended time periods, enabling trend identification and comparison across different time frames.
- **New Insights from Existing Data:** Reanalyzing existing data can uncover new perspectives, trends, and conclusions not identified in the original analysis.
- **Larger Sample Sizes:** Secondary data often comes from large-scale studies, enhancing the reliability and generalizability of findings.
- **User-Friendly:** Secondary data can be utilized by individuals without specialized research skills, making it accessible to a broad audience.



Disadvantages of Secondary Data

- **Not Specific to Your Needs:** Secondary data is collected for purposes other than your own, making it less tailored and potentially less relevant to your specific research or business needs.
- **Lack of Control Over Quality:** You have no control over how the data was collected, which means the accuracy and reliability of secondary data can be questionable and must be critically evaluated.
- **Potential Bias:** Data collected by others may reflect the biases of the original researchers, which could affect its applicability and reliability for your purposes.



too big

too small

Disadvantages of Secondary Data

- **Outdated Information:** Since secondary data was collected in the past, it may not reflect current conditions or recent changes, making it potentially outdated.
- **Non-Proprietary:** Secondary data is often publicly available, meaning competitors can access the same information, providing no unique competitive advantage.
- **Limited Relevance:** May lack the specific variables or detailed information needed for your particular analysis, limiting its usefulness in some cases.



Public Sources of (Secondary) Data

- There are many free public sources of data available on the web.
- As we have seen before, these may not always be a perfect fit for your needs, or it may provide very useful information

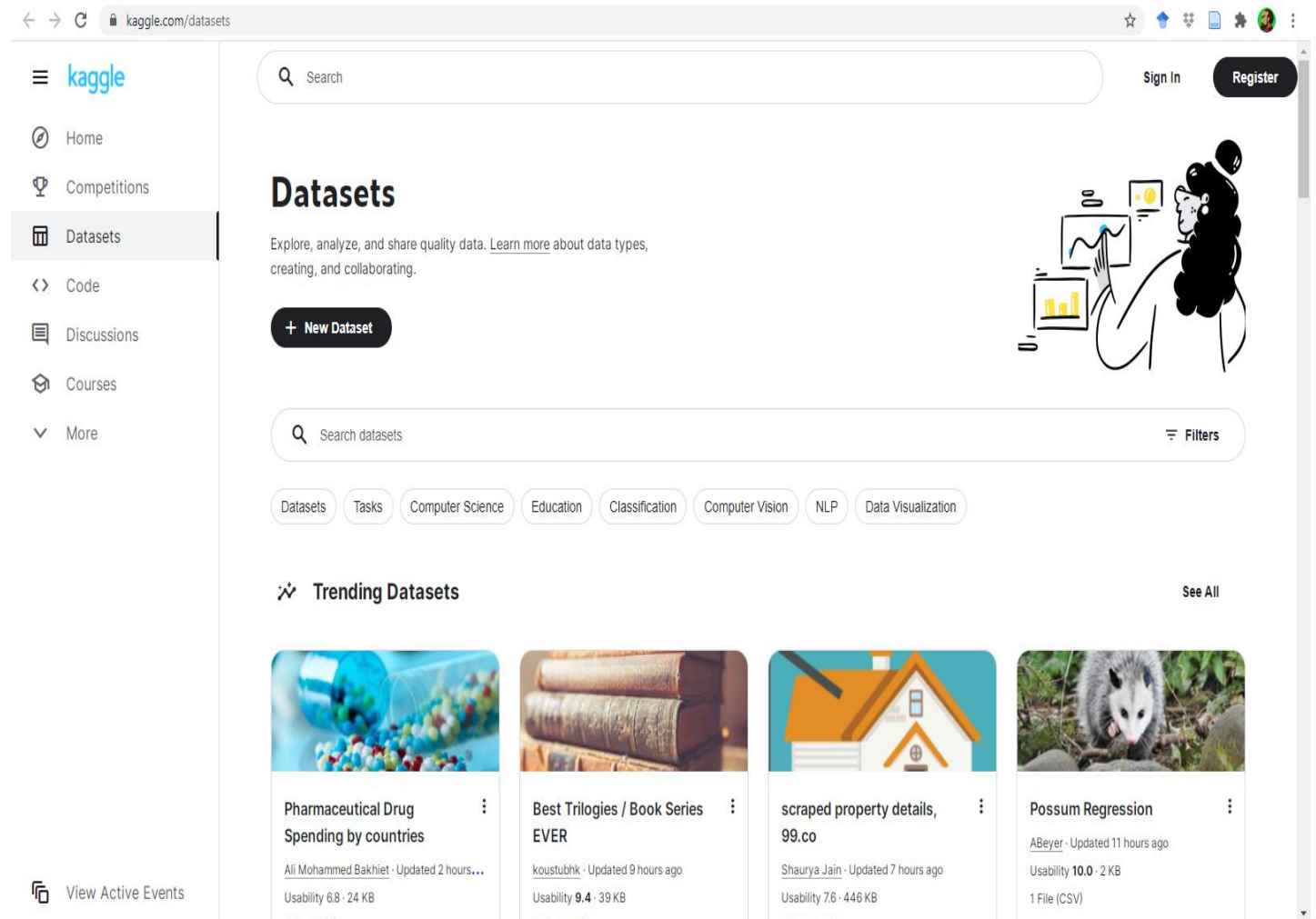
Learning Platform: Offers datasets, tutorials, and courses for all skill levels in data analysis and machine learning.

Competitions: Solve real-world problems, gain practical experience, and build your portfolio.

Extensive Resources: Access thousands of [datasets](#) and coding environments for hands-on practice.

Large Community: Join over 18.5 million users for collaboration, networking, and skill sharing.

Take care! Datasets are uploaded by users, so data reliability may be an issue.



The screenshot shows the Kaggle Datasets page. The left sidebar contains navigation links: Home, Competitions, Datasets (selected), Code, Discussions, Courses, and More. The main content area has a search bar, a 'Sign In' button, and a 'Register' button. Below the search bar is a 'Datasets' section with a description: 'Explore, analyze, and share quality data. Learn more about data types, creating, and collaborating.' There is a '+ New Dataset' button. Below this is another search bar for datasets and a 'Filters' button. A row of category tags includes: Datasets, Tasks, Computer Science, Education, Classification, Computer Vision, NLP, and Data Visualization. The 'Trending Datasets' section is visible, showing four dataset cards: 'Pharmaceutical Drug Spending by countries' by Ali Mohammed Bakhiet, 'Best Trilogies / Book Series EVER' by koustubhk, 'scraped property details, 99.co' by Shaurya Jain, and 'Possum Regression' by ABeyer. Each card displays the dataset title, author, update time, and usability score.

Other sources of free datasets

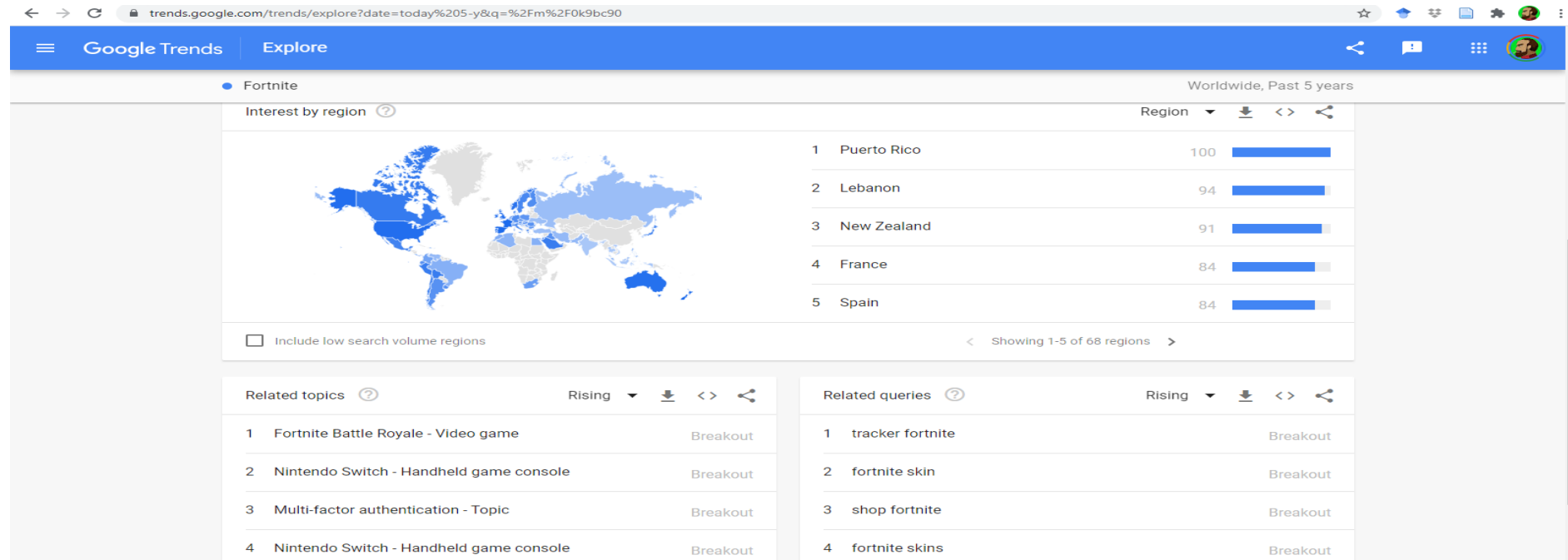
- **[GitHub Awesome Public Datasets](#)**: Repository with a collection of links pointing to datasets.
- **[Google Dataset Search](#)**: A search engine specifically for datasets from across the web, including public data repositories.
- **[70 Free Data Sources](#)**: From Kdnuggets, one of the leading blogs in Data Analytics
- **[AWS Amazon Data Exchange](#)**: Third-party data through Amazon's platform
- **[Instituto Nacional de Estadística](#)**:
- **[UCI Machine Learning Repository](#)**: Provides access to a variety of datasets for machine learning research.
- **Government Data Portals**: Platforms like [data.gov](#) (U.S.) and [data.europa.eu](#) (Europe) offer access to public government datasets.
- **Open Data Platforms**: Websites like [Data World](#) and [FiveThirtyEight](#) provide access to diverse datasets across many fields.
- **[World Bank Open Data](#)**: A resource for global development data, including statistics on economics, health, and education.
- **[UN Data](#)**: Offers datasets related to global issues like population, economy, and environment.

Google Trends

Free tool that analyzes the popularity of search queries in Google Search across various regions and languages, helping users **understand trends, seasonal patterns, and geographic interest.**

Example: Using Google Trends for Analyzing Fortnite Popularity

- **Scenario:** Analyzing the popularity of Fortnite for marketing strategy.
- **Data Collection with Google Trends:**
 - **Tracking Interest:** Analyze search interest in "Fortnite" over time to identify peaks, such as during major in-game events.
 - **Seasonal Patterns:** Spot spike patterns, such as during school holidays or after updates, to optimize marketing campaigns.
 - **Geographic Insights:** Determine regions with the highest search interest to target specific markets.



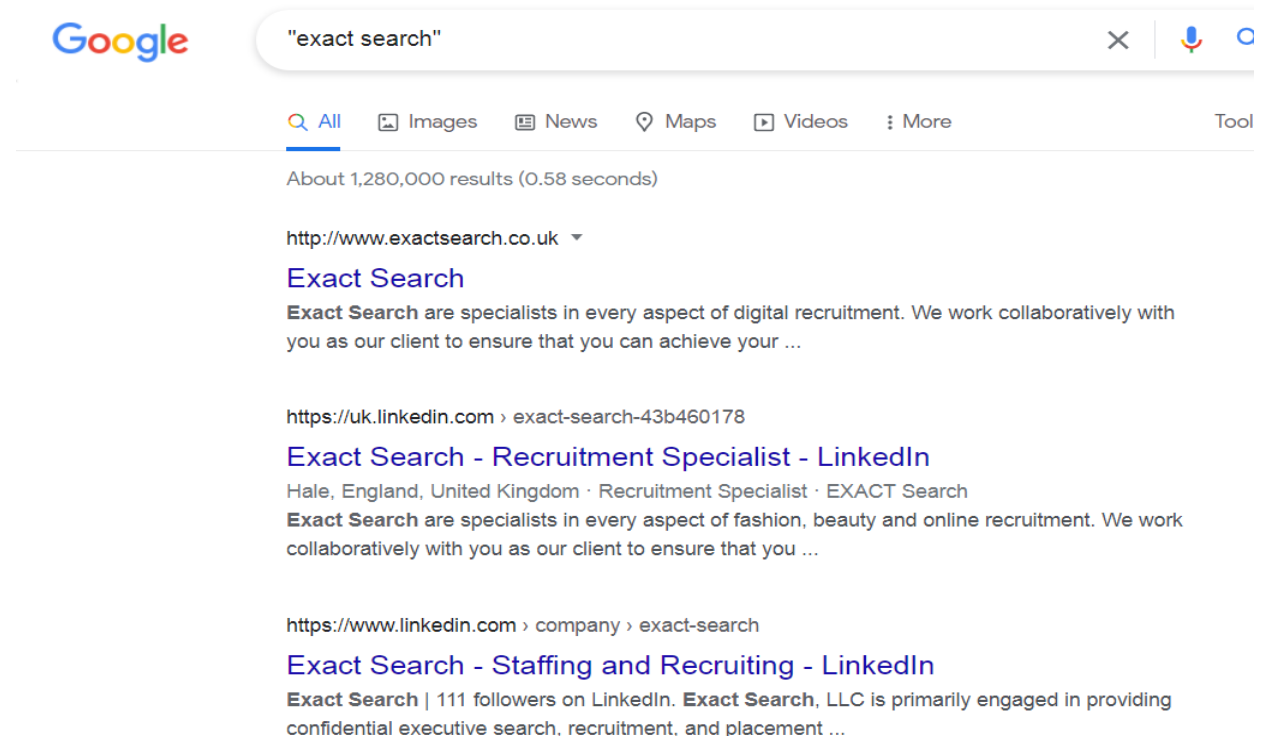
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How to use Google Search Engines to look for information (Advanced Edition)

Google has advanced search capabilities, not displayed by default.

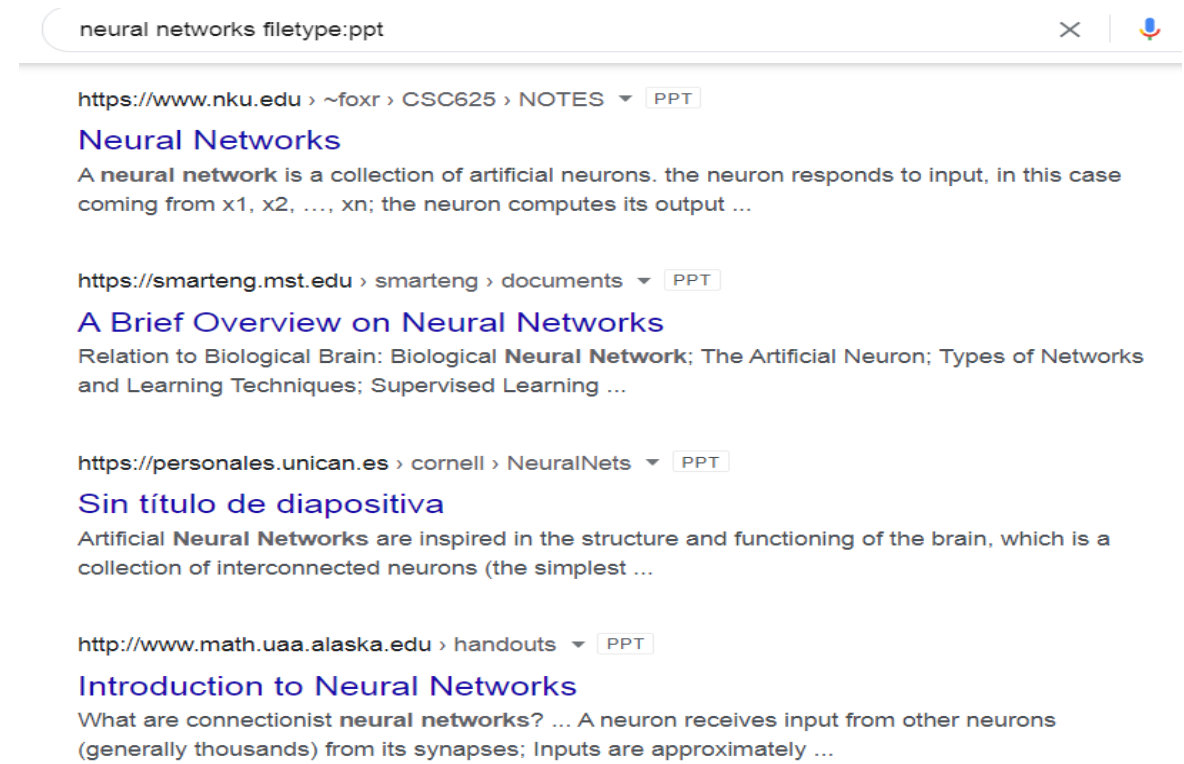
- **Exact match:** Surrounding the search terms with quotation marks tells google Google that the words must appear as an exact phrase.



How to use Google Search Engines to look for information (Advanced Edition)

Google has advanced search capabilities, not displayed by default.

- **Specific format:** To search for a particular file type (pdf, ppt, ect.), add “filetype:”file-extension (e.g., filetype:”pdf)



How to use Google Search Engines to look for information (Advanced Edition)

- **Excluding a term:** To exclude a term, add “-term”

salsa recipe

× | 🔊 🔍

🔍 All 🖼️ Images 📺 Videos 📰 News 🛒 Shopping ⋮ More Tools

About 261,000,000 results (0.69 seconds)

Recipes



Homemade Salsa

Cooking Classy

5.0 ★★★★★ (132)

5 min

Red onion, petite diced tomatoes, lime juice, garlic, chili



Pico de Gallo (Fresh Salsa)

Simply Recipes

4.3 ★★★★★ (113)

15 min

Ribs, fresh tomatoes, lime, red onion, your fingers



Ultimate tomato salsa

BBC Good Food

5.0 ★★★★★ (62)

5 min

White wine vinegar, lime, red onion, tomatoes, garlic

salsa recipe -lime

× | 🔊 🔍

🔍 All 🖼️ Images 📺 Videos 📰 News 🛒 Shopping ⋮ More Tools

About 262,000,000 results (0.64 seconds)

Recipes



Homemade Salsa | 5 Minute Tomato Salsa

Dassana's Veg Recipes

4.8 ★★★★★ (4)

5 min

Lemon juice, cayenne pepper, serrano peppers, chopped



Best Chunky Salsa Recipe

- The Fiery Vegetarian

4.6 ★★★★★ (9)

25 min

Hot sauce, green bell pepper, wine, crushed tomatoes, green



Wonderful Salsa

Food.com

5.0 ★★★★★ (536)

1 hr 15 min

Tomato sauce, green peppers, tomato paste, tomatoes, vinegar

⌵ Show more

How to use Google Search Engines to look for information (Advanced Edition)

The complete list of advanced Google search operators can be found here:

[Google Search
Operators
Cheat Sheet](#)

How to use Google Search Engines to look for information (Advanced Edition)

Google Advanced Image Search

Allows users to refine their search results by specifying criteria like image size, color, type, usage rights, and the exact website, enabling more precise and relevant image searches.

Advanced Image Search

Find images with...	To do this in the search box.
all these words:	Type the important words: winter hoarfrost
this exact word or phrase:	Put exact words in quotes: "frost flower"
any of these words:	Type OR between all the words you want: trees OR weeds OR grasses
none of these words:	Put a minus sign just before words that you don't want: -windows

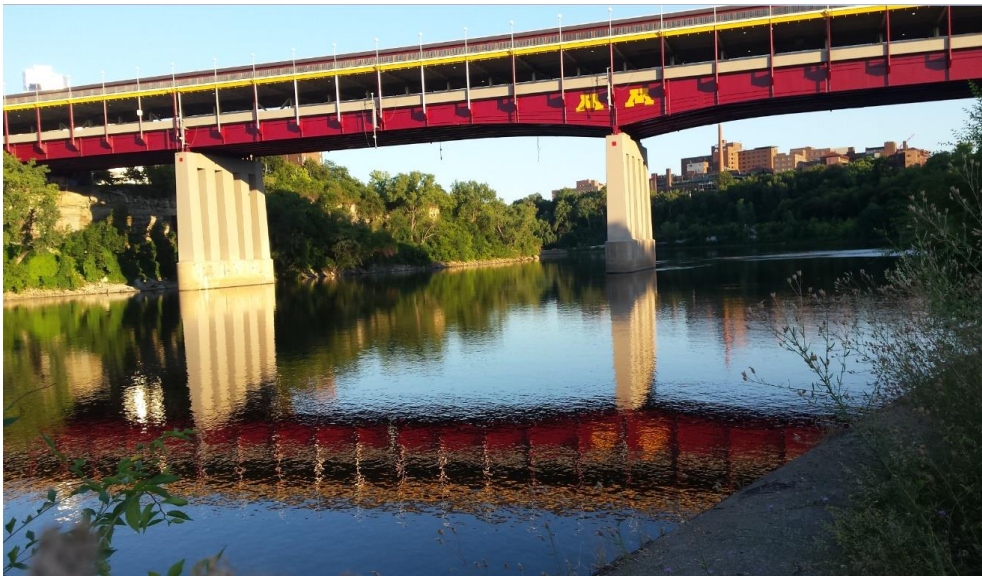
Then narrow your results by...

image size:	any size	Find images in any size you need.
aspect ratio:	any aspect ratio	Specify the shape of images.
colours in the image:	☒ any colour ☐ full colour ☐ black & white ☐ transparent ☐ this colour: 	Find images in your preferred colours.
type of image:	any type	Limit the kind of images that you find.
region:	any region	Find images published in a particular region.

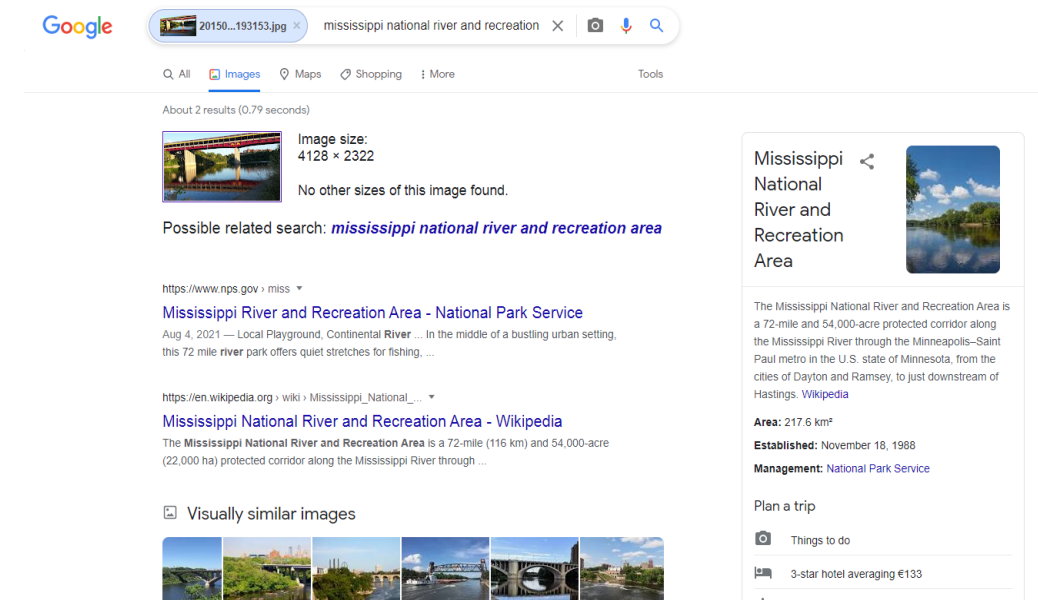
 nn.ppt ^

How to use Google Search Engines to look for information (Advanced Edition)

Reverse Image Search: allows users to upload an image or input an image URL to find information about the image, including its source, similar images, and the contexts in which it appears online, making it useful for identifying objects, verifying image authenticity, and discovering related content.



Picture that I took from the bridge separating the East and West Campuses at the University of Minnesota.



Reverse image Google search result

How to use Google Search Engines to look for information (Advanced Edition)

[Google Scholar](#): search engine that indexes scholarly articles, theses, books, conference papers, and patents from academic publishers, universities, and other scholarly websites, allowing users to easily find and access academic research, citations, and related literature across various disciplines.

Google Scholar

☒ Articles ☐ Case law

Stand on the shoulders of giants



Any time

Since 2021

Since 2020

Since 2017

Custom range...

Sort by relevance

Sort by date

☐ include patents☒ include citations☒ Create alert**Corruption**[A Shleifer](#), RW Vishny - *The quarterly journal of economics*, 1993 - [academic.oup.com](#)

This paper presents two propositions about corruption. First, the structure of government institutions and of the political process are very important determinants of the level of corruption. In particular, weak governments that do not control their agencies experience ...

☆ Cited by 7605 [Related articles](#) [All 25 versions](#)[\[PDF\] jstor.org](#)**Corruption: A review**[AK Jain](#) - *Journal of economic surveys*, 2001 - [Wiley Online Library](#)

As is increasingly recognised in academic literature and by international organisations, **corruption** acts as a major deterrent to growth and development. The aim of this survey is to organise and summarise existing theoretical and empirical work on **corruption** with a view to ...

☆ Cited by 2070 [Related articles](#) [All 8 versions](#)[\[PDF\] wiley.com](#)**[book] Controlling corruption**[R Klitgaard](#) - 1988 - [degruyter.com](#)

Yet too little is said about how to reduce **corruption**. Typical is a recent article, which contains models of corrupt behavior and its spread. At the end of the article, the author says: "Understanding the microeconomic basis of **corruption** is the aim of this paper, not ...

☆ Cited by 4222 [Related articles](#) [All 7 versions](#)**Corruption and growth**[P Mauro](#) - *The quarterly journal of economics*, 1995 - [academic.oup.com](#)

This paper analyzes a newly assembled data set consisting of subjective indices of **corruption**, the amount of red tape, the efficiency of the judicial system, and various categories of political stability for a cross section of countries. **Corruption** is found to lower ...

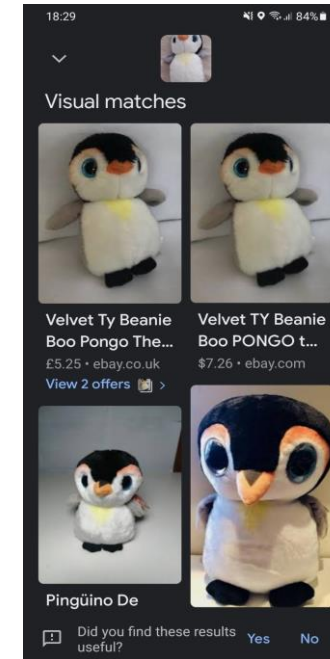
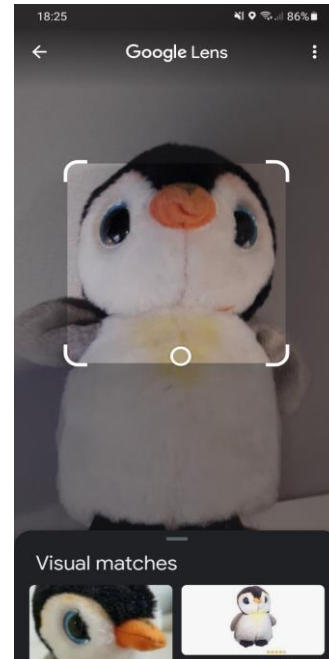
☆ Cited by 12423 [Related articles](#) [All 23 versions](#)[\[PDF\] jstor.org](#)

Related searches

police corruption**eradicating** corruption**bribery** corruptioncorruption **public participation****transparency** corruption**fifa** corruption**fighting** corruption**combating** corruption

How to use Google Search Engines to look for information (Advanced Edition)

Google Lens: AI-powered tool that allows users to search and interact with the world through their camera or images by identifying objects, translating text, scanning QR codes, and providing information or related actions based on what it recognizes, making it useful for real-time information gathering and visual discovery.



Alternative Search Engines

- **Microsoft Bing:**
 - Integrated with Microsoft 365.
 - AI chat features.
 - 2nd largest after Google.
- **Ecosia:**
 - Plants trees with searches.
 - No personal data tracking.
- **DuckDuckGo:**
 - Focus on privacy; no tracking.
- **Qwant:**
 - European-based, privacy-focused.
 - Child-friendly search option.
- **You.com:**
 - Customizable with upvotes/downvotes.
 - Includes YouChat, YouWrite, YouImagine.
- **Perplexity:**
 - AI-driven search engine.
- **Yandex:**
 - Leading in Russia; unique perspective.
- **Baidu:**
 - Top in China; offers different insights.
- **Others:** Yahoo! Search, Brave, etc.
- **Controversial Engines:** ...

How to import data from the Web to Excel

- You can **import data** in tables directly from a web page to **Excel** (no need for Copy+Paste!).
- Example:**
 - Go to the Wikipedia page “[List of countries by alcohol consumption per capita](#)”
 - Open Excel. Go to the Data tab, then select “Import from the Web”
 - Enter the Wikipedia page URL
 - Check the data is correct (sometimes the first row gets duplicated)

Comprehensive list with data from 2016 [edit]

The table below for 189 countries uses 2016 data from the WHO report published in 2018. The methodology used by the WHO calculated use by persons 15 years of age or older. All data in columns refer to year 2016. The column "recorded" refers to the average recorded consumption for the period 2010. Unrecorded consumption (homebrew, moonshine, smuggled alcohol, surrogate alcohol etc.) was calculated using expert judgements and surveys. Total is the sum of the recorded and unrecorded consumption. The next four columns are a breakdown of the recorded alcohol consumption by type. Beer refers to malt beer, wine refers to grape wine, spirits refers to all distilled beverages such as vodka and similar products, and the column "other" refers to all other alcoholic beverages, such as rice wine, soju, sake, mead, kumis, cider, kvass, and African beers (kumi kumi, kwete, banana beer, millet beer, umqombothi etc.). Worldwide consumption in 2016 was equal to 6.4 litres of pure alcohol consumed per person aged 15 years or older.

Pure alcohol consumption among persons (age 15+) in litres per capita per year, 2016^[9]

	Country	Total	Recorded consumption	Unrecorded consumption	Beer (%)	Wine (%)	Spirits (%)	Other (%)	2020 projection	2025 projection
1	Estonia	16.9	15.8	1.1	32.7	7.4	50.3	9.6	11.5	1
2	Lithuania	15	13.8	1.2	43.6	7.3	37.1	12.1	14.4	1
3	Czech Republic	14.3	12.4	1.4	53.3	21.3	25.4	0	11.2	1



Autoguardado Libro1 - Excel

Archivo Inicio Insertar Disposición de página Fórmulas Datos Revisar Vista Ayuda Power Pivot Diseñ

Obtener datos De texto/CSV De la web De una tabla o rango Fuentes recientes Conexiones existentes Consultas y conexiones Consultas y conexiones Actualizar todo Editar vínculos Tipos de datos Cotizaciones Monedas Tipo de datos New Group

A1 Country

	A	B	C	D	E	F	G
	Country	Total	Recorded consumption	Unrecorded consumption	Beer (%)	Wine (%)	Spirits
1	Country	Total	Recorded consumption	Unrecorded consumption	Beer (%)	Wine (%)	Spirits
2	Estonia	16.9	15.8	1.1	32.7	7.4	50.3
3	Lithuania	15	13.8	1.2	43.6	7.3	37.1
4	Czech Republic	14.3	12.4	1.4	53.3	21.3	25.4
5	Seychelles	13.8	12.4	1.4	68.9	22.4	6.3
6	Germany	13.4	11.3	1.4	52.6	28.4	18.9
7	Nigeria	13.4	9.6	3.8	7.9	0.4	0.6
8	Ireland	13	11.3	1.4	47	28	18.8
9	Moldova	13	11.5	1.4	35.4	44.6	20
10	Latvia	12.9	11.1	1.9	42.8	11.1	40
11	Bulgaria	12.7	11.4	1.3	38.8	17.2	42.9
12	France	12.6	11.8	1.5	18.8	58.8	20.7
13	Romania	12.6	10.4	2.2	55.6	28.1	16.4
14	Slovenia	12.6	10.8	1.8	41.4	50.6	8
15	Portugal	12.3	10.6	2.1	26.1	61.5	7.7

Takeaways for the Week – Data Collection Topic

- The importance of having a strategy for your data.
- The different types of Data.
- Structured Data: Pros and Cons.
- Unstructured Data: Pros and Cons.
- Primary vs Secondary Data: Advantages and disadvantages of each.
- Public repositories and directories of datasets.

Skills you Acquired this Week – Data Collection Topic

- Where to find free datasets to use in your data projects.
- How to use Google Trends to find information about the popularity of a Term overtime.
- Advanced use of search engines to find useful information.

